

BENOUALI ABDELKADER YAHIA ZAKARIA

AI, ML & Software Engineering Intern | Master's Student in AI & Cybersecurity

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PROFESSIONAL SUMMARY

High-achieving Master's student in Artificial Intelligence & Cybersecurity (Bachelor's in CS, Top 3 Cohort Rank) seeking a 1–3 month **Remote Internship** in **Artificial Intelligence, Machine Learning, or Software Engineering** between **February 1 and May 31, 2027**. Hands-on expertise building end-to-end Python AI pipelines, LLM integration (Groq API, Gemini 1.5 Flash), Scikit-learn ML models (98% accuracy), RESTful API backends (Node.js/Express, Nuxt 3), and containerized cloud deployments (Docker, PostgreSQL, Render, 88 production builds). Proven global remote collaboration skills through Harvard-affiliated leadership training (Aspire Leaders) and international exchange (Erasmus+). Eager to contribute to production AI agents, RAG workflows, backend engineering, and MLOps pipelines.

TECHNICAL SKILLS

AI, ML & NLP: Artificial Intelligence, Machine Learning (Scikit-learn, SVM, Random Forest), Large Language Models (LLMs), Natural Language Processing (NLP), Prompt Engineering, RAG Concepts, Generative AI, AI Ethics

AI Agents & Automation: AI Workflows, Process Automation, Dynamic AI Agents, Data Pipeline Engineering

Programming Languages: Python, JavaScript, TypeScript, C Programming, HTML5, CSS3, SCSS

Backend, APIs & Databases: Node.js, Express.js, RESTful APIs, Groq API, Google Gemini API, Nuxt.js 3, MeiliSearch, PostgreSQL, MongoDB

Cloud, MLOps & Tools: Docker, Render Cloud Platform, Git/GitHub, Linux (Ubuntu), Streamlit, Windows Administration

UI/UX & Prototyping: Figma, UI/UX Design, Glassmorphism UI, Responsive Web Layouts, Wireframing

Languages: Arabic (Native), English (Professional Working / Academic B1/B2), French (Professional Working B1)

EDUCATION

Master's Degree in Computer and Information Sciences — University Ain Temouchent | September 2025 – July 2027 (In Progress), Ain Témouchent, Algeria

- **Specialization:** Artificial Intelligence & Cybersecurity.
- **Core Coursework:** Artificial Intelligence, Machine Learning, Deep Learning, Natural Language Processing, Cybersecurity Fundamentals, IT Security, Data Ethics, Information Systems.

Bachelor's Degree in Computer Science — University Ain Temouchent | September 2022 – June 2025, Ain Témouchent, Algeria

- **Academic Distinction:** Ranked **Top 3 (3rd place)** out of the entire cohort across the first 4 semesters (L1 & L2).
- **Core Competencies:** C Programming, Data Structures, Algorithms, Computer Networking (TCP/IP), Operating Systems.

FEATURED AI, ML & SOFTWARE ENGINEERING PROJECTS

Adaptive Blended Assessment (WriteLens V2) — 3-Tier AI Platform & MLOps Pipeline | Python, Groq LLM API, Node.js, Express, React, TypeScript, Docker, PostgreSQL, Render

- Architected a 3-tier academic AI analytics platform processing student text through a custom Python AI engine (WriteLens V2), generating diagnostic profiles, risk bands, and cohort intervention summaries.
- Engineered a privacy-first LLM pipeline integrating Groq API with strict mode tokenization, fail-closed security, and provenance tracking across 88 production deployments on Render Cloud.
- Built a REST API backend (Node.js/Express) with run bundle archival endpoints (/api/writelens/runs) and containerized the architecture using Docker and render.yaml CI/CD configurations.

EcoSentinel AI — Environmental Water Quality Monitoring System | Python, Scikit-learn, SVM (RBF Kernel), Pandas, NumPy, Streamlit, Glassmorphism UI

- Engineered an aquatic pollution detection pipeline utilizing Scikit-learn SVM (RBF kernel) achieving **98% accuracy** in classifying water quality from sensor data (turbidity, dissolved oxygen).
- Built an interactive Streamlit web dashboard featuring real-time parameter sliders, instant SVM predictions, and bilingual Arabic/French UI support.
- Authored a formal academic LaTeX research report (rapport_sentinelle_ecologique.pdf) documenting dataset generation, feature engineering, and model evaluation.

NewsBot — AI-Powered News Assistant & NLP Chatbot | *Node.js, Express, Google Gemini 1.5 Flash API, JavaScript, REST API, HTML5/CSS3*

- Designed and deployed an AI-powered news chatbot using Google Gemini 1.5 Flash API and Express.js, providing real-time news summarization and NLP classification across 8 news domains.
- Crafted specialized system prompts for news analysis and fact-checking assistance; implemented input validation, secure environment key management (`.env`), and application logging (`combined.log`, `error.log`).

PFC (Plateforme de Fin de Cycle) — Full-Stack Graduation Web Application | *Nuxt.js 3, Vue 3, TypeScript, MongoDB, MeiliSearch, JWT Auth, REST APIs*

- Engineered a full-stack platform incorporating Nuxt.js 3, TypeScript, MongoDB, and MeiliSearch for sub-second full-text search across community articles and live news feeds.
- Implemented JWT role-based authentication (RBAC), content moderation APIs, Gmail SMTP contact integration, and an embedded AI ChatBot component.

Wanderlust AI — Conversational AI Travel Planning SPA | *JavaScript (ES6+), Google Gemini API, Tailwind CSS, REST APIs*

- Engineered a Single Page Application (SPA) integrating Google Gemini API for natural language itinerary generation alongside 5 external APIs (YouTube, Weather, Search, Pexels).

EXPERIENCE & INTERNATIONAL PROGRAMS

Participant — Aspire Leaders Program — Aspire Institute (Harvard-affiliated) | July 2026 – Present, Remote Global

- Selected for Cohort 3 of a competitive global leadership development program for high-potential international leaders.
- Attended live masterclasses with Harvard educators (including Michael Horn) on career growth, technical innovation, and strategic problem-solving.
- Collaborated in cross-cultural remote teams on real-world case studies, refining communication and professional leadership skills.

Participant — Erasmus+ Virtual Exchange Program — VIRTUALLYEDU | July 2026 – Present, Remote International

- Engaged in an international Erasmus+-funded program focused on digital skills, cybersecurity fundamentals, and cross-border virtual teamwork.

Participant & Developer — UNESCO Youth Hackathon 2026 — UNESCO | 2026, Remote International

- Developed **EchoBreaker**, an interactive "Algorithm Simulator" web app visualizing social media recommendation algorithms and echo chamber effects in real time.
- Engineered educational tools and algorithmic bias mitigation toolkits promoting digital literacy, aligned with UNESCO's Sustainable Development Goals (SDGs).

CERTIFICATIONS & TRAINING

- **Google AI Essentials** — Google (Generative AI, Data Ethics, Responsible AI Fundamentals)
- **Crash Course on Python** — Google via Coursera (Credential ID: CLGGTK62WSHC)
- **Advanced Computing Track** — Algerian-American Summer University 2025 (7-day intensive program in AI, ML, Cybersecurity, and AI Ethics with US/Algerian professors)
- **Google IT Support Professional Certificate** — Google via Coursera (Credential ID: GCE48W62HKVF)
- **Microsoft 365 Fundamentals Specialization** — Microsoft via Coursera (Credential ID: HBNELWZMCG63)